

Computer Workshop  
Assignment 3: Core Git

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**Assignment Notes:**

1. Some questions may include material not yet covered in class. These questions are meant to help you develop your research skills.
2. homework is strictly forbidden. If caught, both students involved will receive a score of 0 on this assignment
3. If you have any questions, feel free to ask in the course's Telegram group to benefit your fellow classmates.
4. It's recommended to visit Git's official documentation for additional information on its commands.

## 1 More Git commands

### 1.1 Cloning

In this section, you will be learning about a core feature in Git called cloning. read about cloning and answer the following questions:

1. What does cloning a repository mean?
2. What is the command for cloning a GitHub repository?
3. Does cloning a repository retain its commit history?

### 1.2 Forking in Open-Source Projects and GitHub

In this section, you will explore the concept of forking in the context of open-source projects on GitHub. Read about forking and answer the following questions:

1. What does it mean to fork a repository on GitHub?
2. Describe the process of forking a repository.
3. How does forking contribute to collaborative development in open-source projects?
4. After forking a repository, how can you keep your fork up-to-date with the original repository's changes?
5. How can you contribute to the original repository after forking it?

### 1.3 Git Resetting

In this section, you will delve into the Git command `git reset`. Read about Git resetting and answer the following questions:

1. . What is the purpose of the `git reset` command in Git?
2. Describe the difference between `git reset` and `git revert`.
3. Explain the three primary modes of `git reset` (`-soft`, `-mixed`, and `-hard`) and how they affect the staging area and working directory.

## 2 Merging and Rebasing

In this section, you will be given a GitHub URL that contains a repository that you need to fork and make your own necessary changes on them and push them to your own GitHub repository.

RepositoryURL: <https://github.com/MiliAxe/cw1402.proj/tree/master>

Follow the necessary instructions on the projects given below.

### 2.1 First project

In this project, you are given a project that consists of three branches called master (main branch), feature-one and bug-fix. You have to first merge bug-fix into feature-one and after that, merge the branch feature-one into master

Here is a demonstration of how this project looks like:

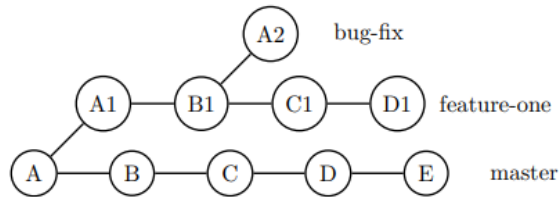


Figure 1: Project 1

After merging the branches, your project should look like this:

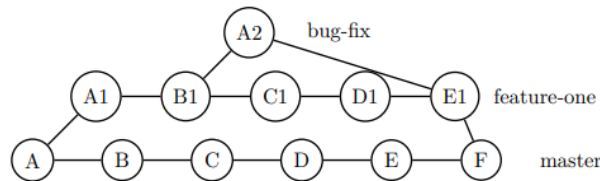


Figure 2: Prpject 1 after merging

**NOTE:** In case of any conflicts, keep the content of both of the branches.

### 2.2 Second project

In this project, you are given a project that consists of three branches called master (main branch), feature-one and bug-fix. You have to rebase bug-fix into feature-one and after that, rebase the branch feature-one into master.

Here is a demonstration of how this project looks like after and before re-basing:

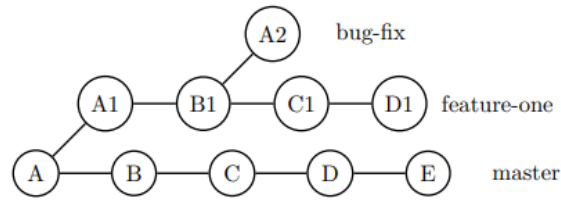


Figure 3: Project 1

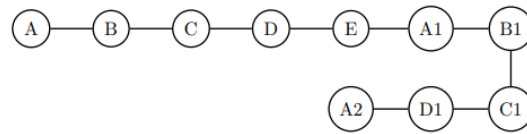


Figure 4: Project 2 after rebasing

**NOTE:** In case of any conflicts, keep the content of both of the branches.

**QUESTION:** Is it possible to have two forks of the same repository on your GitHub account? If yes, how can you do that? If no, provide a solution using cloning and adding a remote repository.

## 2.3 Instructions

For better understanding of what should be done in this section, here are some instructions:

1. Head to the GitHub URL provided
2. Fork the repository provided and rename the repository to the format HW\_STUDENTNO
3. Apply the necessary changes stated in the sections.
4. Push the changes to your forked repository
5. Provide the URL to your GitHub repository in your assignment's PDF as well as the commands you applied on the repository.

**NOTE:** You should provide two URLs in your assignment's PDF, one for each project.



Figure 5: picture of test

### 3 the end of the project